

GPU Computing

Exercise 5

Luis Wenzel
Miro Joensuu
Daniel Bayer
Noah Schneider

1 Presenting

Willing to present:

6.1

6.2 Yes

6.3

6.4

6.5

6.6

2 Reading

3 CPU reduction

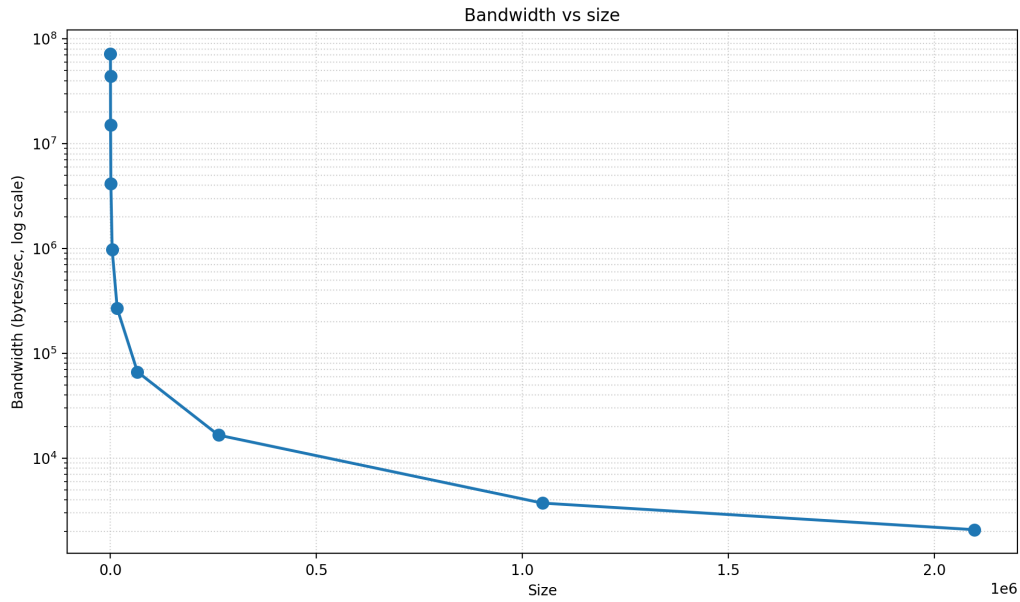


Figure 1: CPU: Bandwidth vs. size of data

1 shows the bandwidth of the kernel of the naive CPU implementation. Measurements are done with an average over 10 runs. What can be observed is that the bandwidth decreases with increasing data sizes. This is due to the fact that, with data getting larger, higher and thus slower levels of the cache hierarchy need to be used.

4 GPU reduction: naive

5 GPU reduction: parallel

6 GPU reduction: parallel + volta

7 GPU reduction: profiling